

M-Line

Multifunctional masterpieces



))) M-Line

Great performance and small size

The **M-Line** consist of self-powered modular PA systems suitable for in- and outdoor audiences ranging from small crowds to hundreds of people.

When great sound matters, but space for placing is scarce or transportation needs to be made easier, you will benefit from the solutions offered by this flexible, compact and sophisticated product line. Every component is exclusively designed and made by SE Audiotechnik.

All units are equipped with class-D amplifiers combined with rock-solid DSP filtering and limiting provide outstanding audio quality, whilst ensuring worry-free driver protection. All products are available in both black and white polyurea coated finishes.



SUB 112BR

SUB 210BP

M-F3A FS

M-F3A



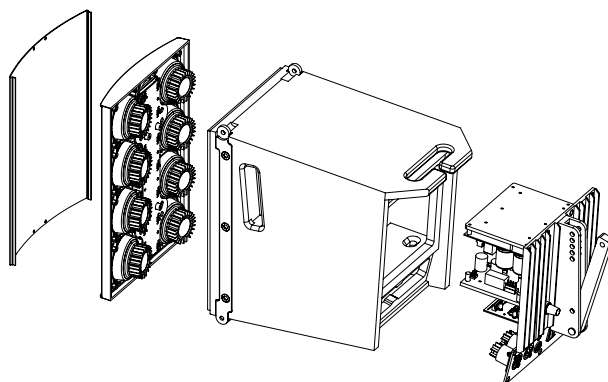
))) M-F3A

A masterpiece of sophisticated German engineering

M-F3A is an ultra-compact active line array system offering crystal-clear sound reproduction with studio quality highs and punchy mids due to its hornless design. The unique PA sound is achieved by incorporating 8 high-efficiency 2.8" neodymium mid-frequency drivers and 7 pieces of 1" direct radiation dome tweeters into each unit.

All drivers are mounted on an A4 paper-sized front surface while weighing only 8 kg including amplification and DSP. Each unit can produce an impressive 115 dB SPL continuously and reaches up to 120 dB peak SPL. 120° horizontal and 16° vertical dispersion angles enable the system to cover vast audience areas.

By using only a single appropriate bumper frame, a unique line assembly of up to 12 speaker boxes can be put into practice. Thanks to a easy-to-use and precise rigging mechanism you have control over each unit's alignment, ensuring optimum coverage and long-throw capabilities. The variety of ground-based and flown subwoofers extends the capabilities of the system to the lowest frequencies.



Specifications

M-F3A

Type	bass-reflex
Frequency Response (-6 dB)	120 Hz - 20 kHz
Amplification	200+200 W RMS, Class-D, SMPS
Max. SPL (1m)	120 dB (128 dB ¹)
Dispersion (H x V)	120° x 16°
Splay Angles	0°, 1°, 2°, 3°, 4°, 6°, 8°
Transducers	MF: 8 x 2.8" drivers with neodymium magnet and cooling module, 0.75" voice coil HF: 7 x 1" with neodymium magnet and cooling module, 1" voice coil
DSP	48 kHz, 24-bit DSP processor with eXtended dynamics processing latency: 1.1 ms
Signal Processing Protection	Multiband-limiter, EQ short-circuit, over-temperature, over-current
Controls	On/Off switch, Main Level, High Shelf (0 dB to +6 dB hybrid design)
Audio Connectors	XLR 3-pin female line-level input, XLR 3-pin male signal link
Power Connectors	Neutrik Powercon® type A/B
Rigging	3-point rigging with SE rigging pins, up to 12 elements with SE bumper frame (flown), up to 4 elements with SE ground stack frame, up to 2 elements with SE U-bracket
Cabinet	12 mm plywood, black or white polyurea coating
Dimensions (W x H x D)	210 x 317 x 359 mm
Weight	8 kg

¹ Measured using EIA tone bursts.



))) **SUB 112BR & SUB 210BP**

A reliable ground support for M-Line

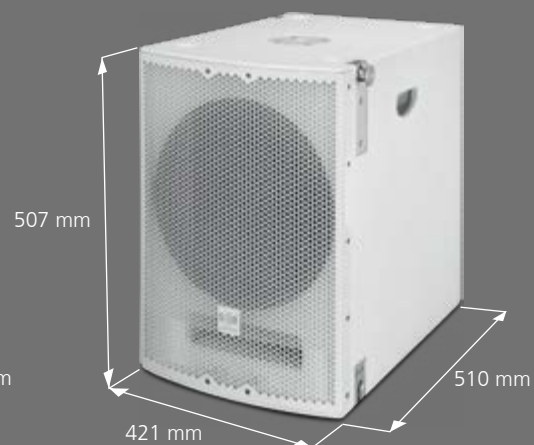
The **SUB 112BR** 12" bass-reflex type, as well as the **SUB 210BP** v-shaped double 10" bandpass type subwoofer add punchy lows to the **M-F3A** system. Both units are powered by an updated built-in 800 watts class-D amplifier.

Their efficient design and compact size guarantees flexible placement in all kinds of environments. The subwoofers provide a two-point SE rigging system, which allows for a safe & easy stacking. Smaller PA systems can also be built with ease by either using a U-bracket with distance pole or an SE stacking frame.

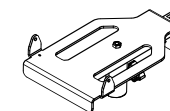


Specifications

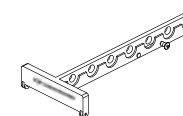
	SUB 210BP	SUB 112BR
Type	V-shaped arrangement, bandpass type	bass-reflex
Frequency Response (-6 dB)	45 Hz - 155 Hz	42 - 250 Hz
Power Output (RMS)	400 W	400 W
Max. SPL (1m, half-space)	121 dB	121 dB
Transducers	2 x 10" drivers with ferrite magnet and 2" voice coil	1 x 12" driver with ferrite magnet and 3" voice coil
Power Amplifier Type	Class-D, SMPS	Class D, SMPS
Input sensitivity	-18 dBV	-18 dBV
DSP	48 kHz, 24-bit DSP processor, processing latency: 1.1 ms	48 kHz, 24-bit DSP processor, processing latency: 1.1 ms
Signal Processing	Limiter, EQ	Limiter, EQ
Protection	Short-circuit, over-temperature, over-current	Short-circuit, over-temperature, over-current
Controls	On/Off switch, Main Level, Low-pass frequency (80-240 Hz), Polarity inversion	On/Off switch, Main Level, Low-pass frequency (80-240 Hz), Polarity inversion
Indicators	Power On, Signal In, Limit, Protect	Power On, Signal In, Limit, Protect
Audio Connectors	XLR 3-pin female line-level input, XLR 3-pin male signal link	XLR 3-pin female line-level input, XLR 3-pin male signal link
Power Connectors	Neutrik Powercon® type A power input, type B power link (MAX: 15A, Maximum 6 pcs per line)	Neutrik Powercon® type A power input, type B power link (MAX: 15A, Maximum 6 pcs per line)
Rigging/Hardware	2x two-point SE stacking system, M20 distance pole thread	2x two-point SE stacking system, M20 distance pole thread
Power Requirements	100 - 120 VAC or 220 - 240 VAC	100 - 120 VAC or 220 - 240 VAC
Cabinet	15mm plywood, black or white polyurea coating	15mm plywood, black or white polyurea coating
Dimensions (W x H x D)	421 x 507 x 510 mm	355 (385 with rigging) x 506 x 496 mm
Weight	28 kgs	22.5 kgs



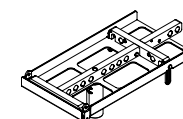
Accessories



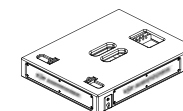
M-F3A UB
U-Bracket



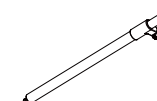
M-F3A BF
Bumper Frame



M-F3A FB
Multi Purpose
Rigging Frame



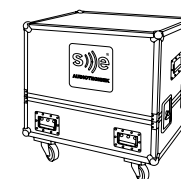
M-F3A SF
Ground Stack
Frame



M-F3A SPS 20
M20 Pole support



M-F3A FC
Flight Case (for 4
pcs.)



M-F3A SUB FC
Subwoofer Flight
Case
(for 2 pcs.)

))) M-F3A FS

The long awaited bass extension

The **M-F3A FS** flysub is a long awaited bass extension module for the well known **M-F3A** line array system. By employing two specially designed 6x9" woofers the subwoofer maintains a slim and discreet design. The height is exactly double the size of two **M-F3A**'s, which complements the overall system dimensions.

A built-in 800W class-D power amplifier and 24-bit DSP guarantees maximum compatibility and performance with M-Line systems. In standalone mode the unit can deliver rich & deep bass to manage simple applications. More demanding applications can be realized by pairing the system with ground-based subwoofers, thus working as a bass extension for low-mid frequencies.

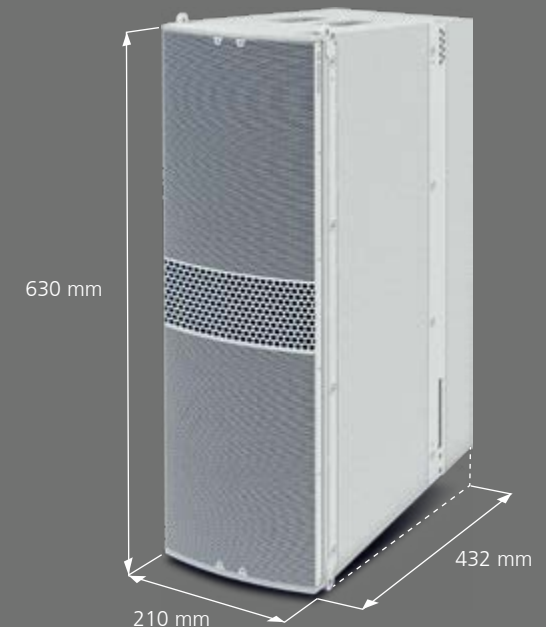
Improved rigging capabilities and a newly designed bumper frame gives additional flexibility to integrate this adaptable subwoofer into **M-F3A** systems.



Preliminary specifications

M-F3A FS

Type	bass-reflex
Frequency Response (-3 dB)	56 - 250 Hz
Frequency Response (-6 dB)	53 - 350 Hz
Amplification	800 W Class-D
Max. SPL	122 dB (119 dB continuous)
Sensitivity (1W / 1m)	93 dB
Dispersion (H x V)	omnidirectional
Transducers	Subwoofer: 2x 6 x 9" woofers
Input	1x Line level (XLR)
Output	1x Line level (XLR)
Indicators	Power on, Signal, Limit, Protect
Controls	1x Main level, 1x LP filter
DSP	48 kHz, 24-bit processor with eXtended dynamics processing, latency: 1.1 ms
Adjustable Lowpass	100 - 350 Hz (-6 dB)
Power Connectors	Neutrik Powercon® Input / Link
Rigging	4-point rigging, 4x standard SE 6 mm pins
Cabinet	Plywood, black or white polyurea coating
Dimensions (W x H x D)	210 x 630 x 432 mm
Weight	21 kg



Application examples



Micro system



For audiences of **up to 100**

- 2x **M-F3A**
Line Array units
- 1x **M-F3A**
U-Bracket
- 1x **M-F3A SUB 210BP**
Subwoofer



Short-Throw Setup

Applications: presentation events, public speech, small venues, cafes, bars, stage fills

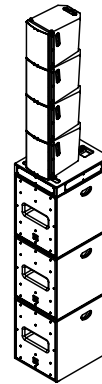


Mini system



For audiences of **up to 200**

- 4x **M-F3A**
Line Array units
- 1x **M-F3A**
Stacking Frame
- 3x **M-F3A SUB 210BP**
Subwoofer



Short- to Medium-Throw Setup

Applications: stage fills, mid-sized touring, clubs, outdoor stages, concert venues

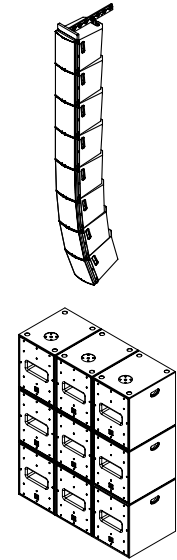


Introduction system



For audiences of **up to 600**

- 1x **M-F3A**
Bumper Frame
- 8x **M-F3A**
Line Array units
- 9x **M-F3A SUB 210BP**
Subwoofer



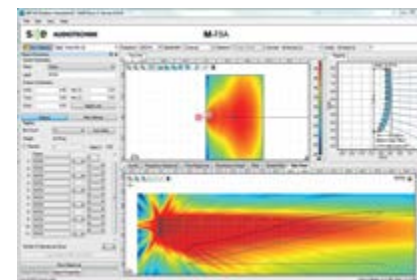
Medium-Throw Setup

Applications: touring, live concerts, small festivals, multipurpose halls, event centers

EASE® & EASE® Focus 3 ready

A set of EASE® *.GLL files is available for all system components including: M-F3A, **SUB 112BR** & **SUB 210BP**. EASE® is an industry standard acoustic simulation software for a wide range of applications and venues, especially for installations in houses of worship, stadiums, performance centers, PA & others.

EASE® Focus 3 is an acoustic prediction software well suited for live sound requirements. Audience coverage, sound pressure levels, rigging and other parameters can be simulated for the **M-F3A** system to ensure good sounding results and faster set-up times.



Focus 3 can be downloaded free of charge on the AFMG website: <http://focus.afmg.eu>

The M-F3A *.GLL files are available both from the AFMG Focus and our SE Audiotechnik website.

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M-Line brochure v. 2.0 | Design by STH design & consulting
All product specifications in this document are subject to change without notice.

M-F3A

Pristine acoustic reproduction

120° horizontal coverage

Ultra-compact size

Built-in amplification & DSP

Low weight of only 8 kgs

